

Understanding Uncertainty in Data-Driven Research

How Uncertainty Shapes Machine Learning



Understanding Uncertainty in Data-Driven Research

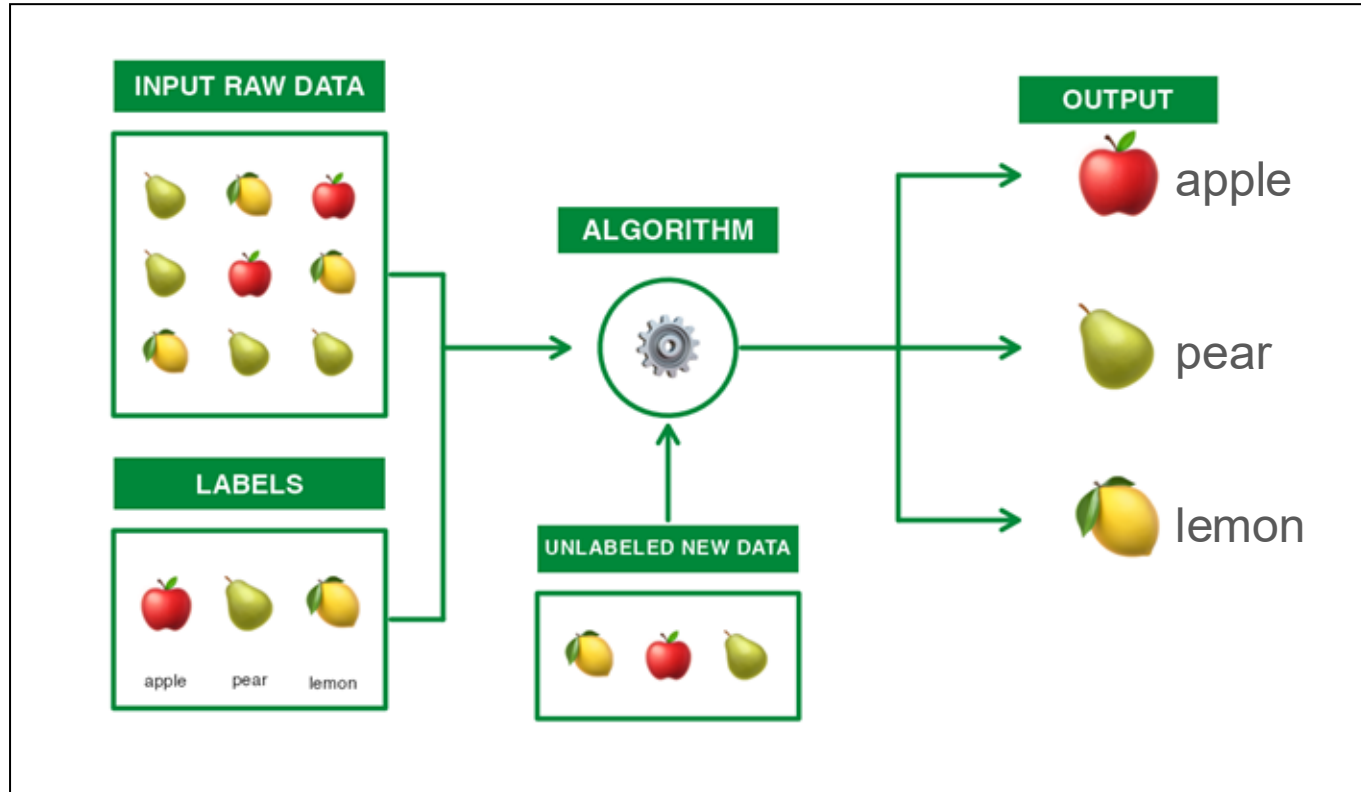
Questioning Ground Truth Labels

What is a label?

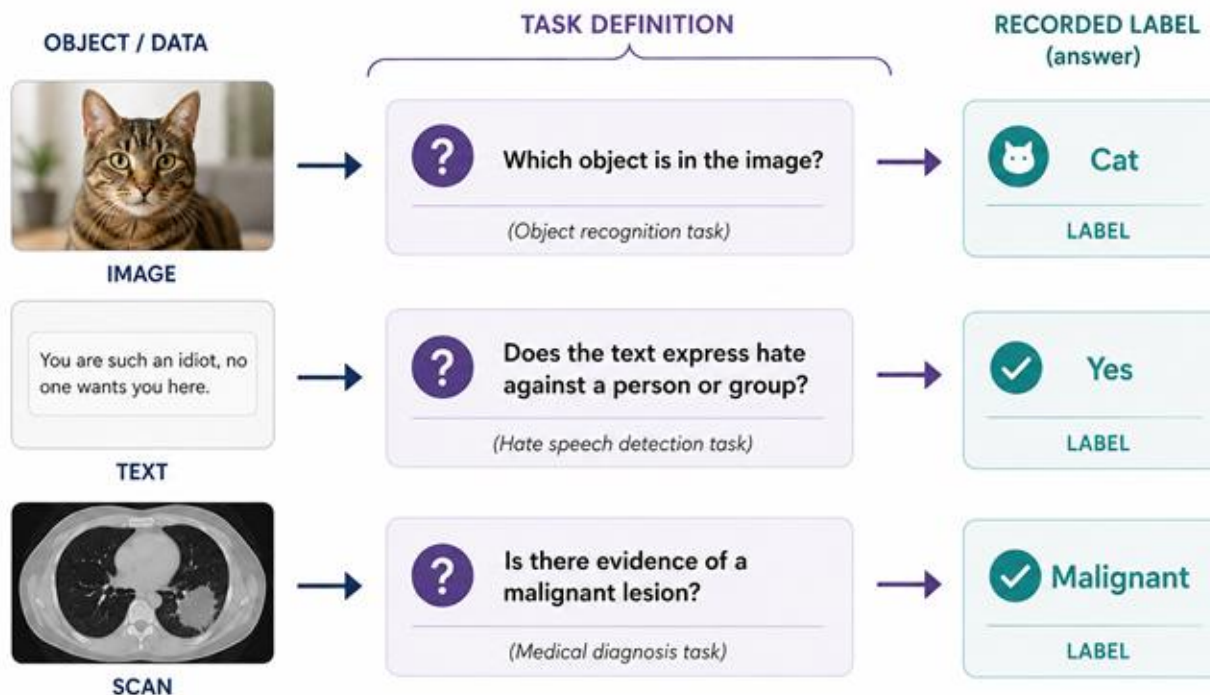
Unit 2



Labels in Supervised Machine Learning



A label is an answer to a question



The label records an answer produced for a defined task.

The same case can support different labels

Consider a social-media post:

Research question	What the label intends to represent
Does it contain an insult?	observable linguistic feature
Does it violate platform policy?	institutional category
Is it illegal hate speech?	legal category
Does the targeted person experience harm?	reported human experience



A label only has meaning in relation to the specific property, judgment, or outcome it is intended to represent.

Why labels vary

Before judgement:

information available to the labeling process

- image or sensor quality
- available evidence
- available context

During judgement:

how the label is assigned

- interpretation
- expertise
- unclear instructions
- attention

After judgement: how labels are recorded and stored

- transcription/coding
- aggregation
- preprocessing or discretization
- changing label definitions

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Understanding Uncertainty in Data-Driven Research

Questioning Ground Truth Labels Agreement Metrics & Majority Vote

Unit 2



Agreement metrics

Percent agreement

$$P_o = \frac{\text{number of matching judgements}}{\text{number of judgements}}$$

fast and easy

does not account for
random guessing

Cohen's κ

Two annotators:

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

P_o : observed agreement

P_e : expected agreement by
chance

Krippendorff's α

Two or more annotators

Allows missing annotations

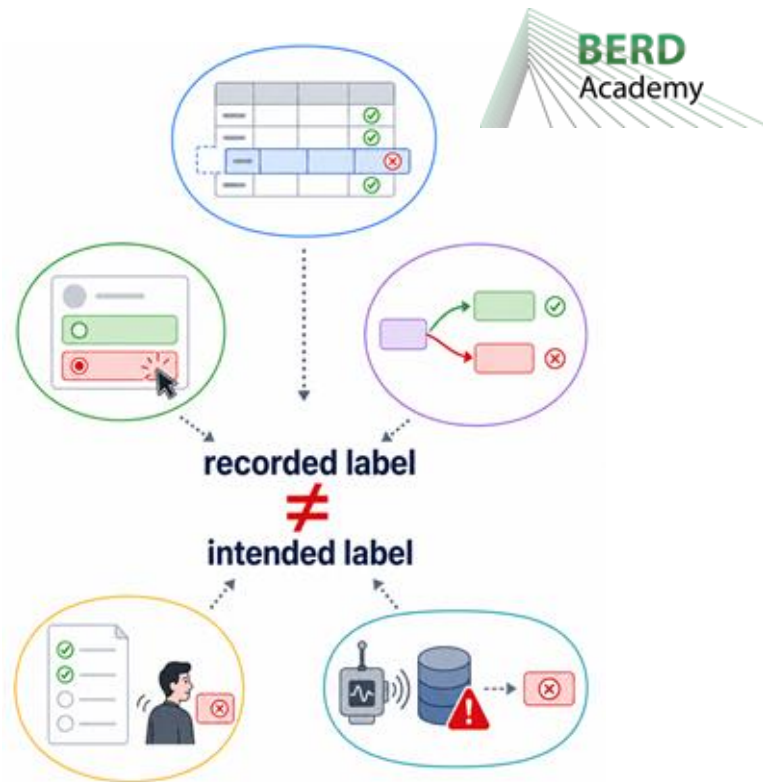
Can be adapted to
nominal, ordinal, interval,
or ratio data

**Agreement measures consistency between recorded judgments,
not whether the labels are correct or scientifically valid.**

Some variation is error

- a row shifted during data entry
- the wrong button clicked
- a code mapped to the wrong category
- an annotator ignored clear instructions
- a sensor or administrative system failed

Here, correction or quality control is appropriate.



Majority vote

Repeated labels can average out independent errors.

If each labeler is **usually right** and their mistakes are sufficiently independent, the **majority can be more reliable than one labeler**.

The majority vote is then taken as **ground truth**.

Assume:

- one correct binary label exists
- each labeler is correct with probability 0.70
- errors are independent



Majority vote
(at least 2 of 3 agree)

Success cases for majority vote

Exactly 2 correct



(3 ways to arrange)

All 3 correct



(1 way)

$$3(0.7)^2(0.3) + (0.7)^3 = 0.784$$



70% individual accuracy
becomes **78.4%** majority accuracy.

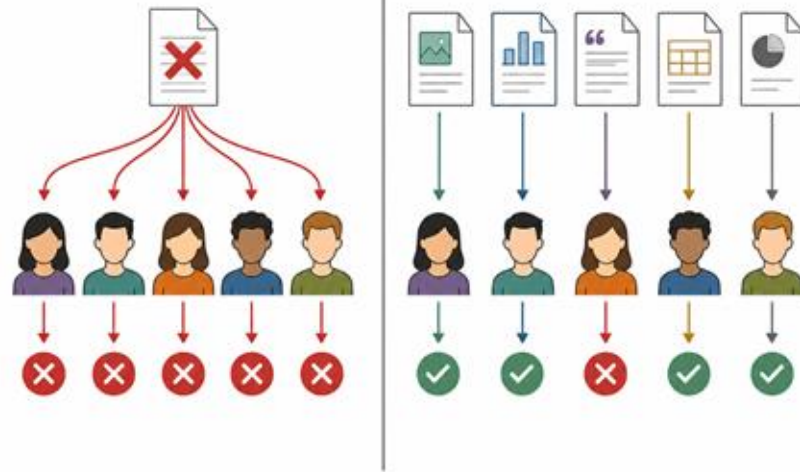
Caveats of majority voting

⚠ Correlated errors do not average out

Labeler vote's are not independent evidence if they

- use the same flawed source,
- share the same blind spot,
- or copy one another.

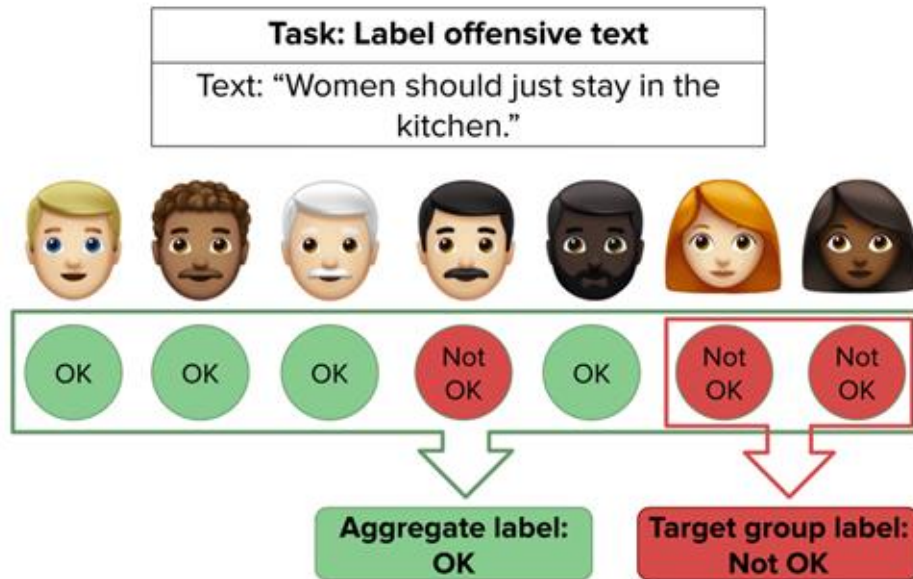
Ten repetitions of one mistake are not ten independent measurements



Majority vote compresses a distribution

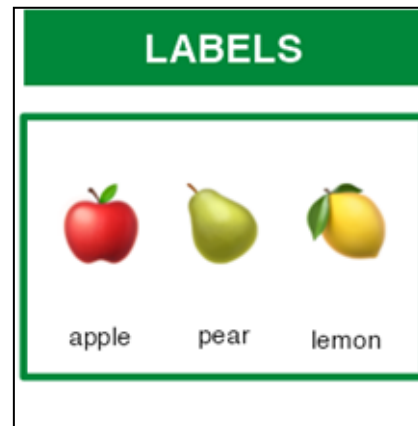
⚠️ A majority can silence a relevant minority.

✅ Disagreement can be a signal.



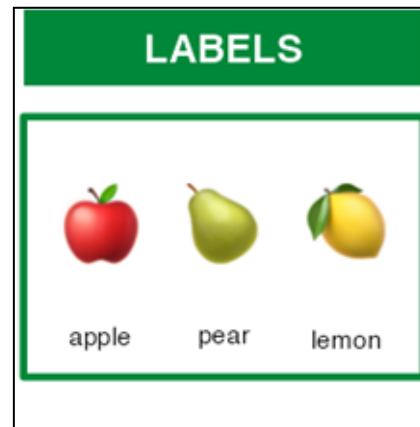
Conclusion

- Supervised learning usually assumes that each training example has **one correct label**
- Majority voting allows us to train and evaluate models with simple targets and metrics



Conclusion

- In practice, labels are produced through measurement, coding, annotation, diagnosis, expert judgment, platform rules, legal definitions, or administrative processes
- A label can be wrong, noisy, ambiguous, incomplete, contested, or valid only for a specific purpose
- "Ground truth" often means "the label we decide to treat as true for this task."



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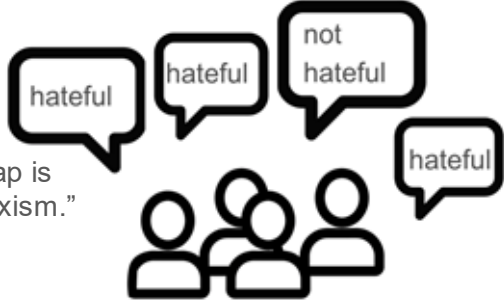
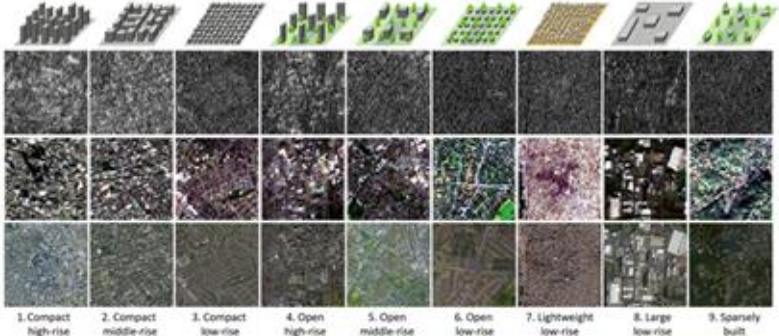
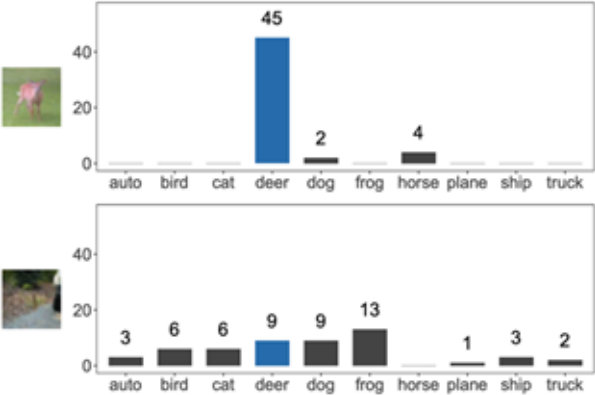
Understanding Uncertainty in Data-Driven Research

Questioning Ground Truth Labels Human Label Variation

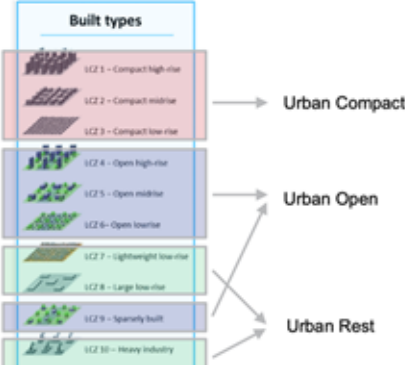
Unit 2



Human Label Variation



“The gender-pay-gap is not a problem of sexism.”



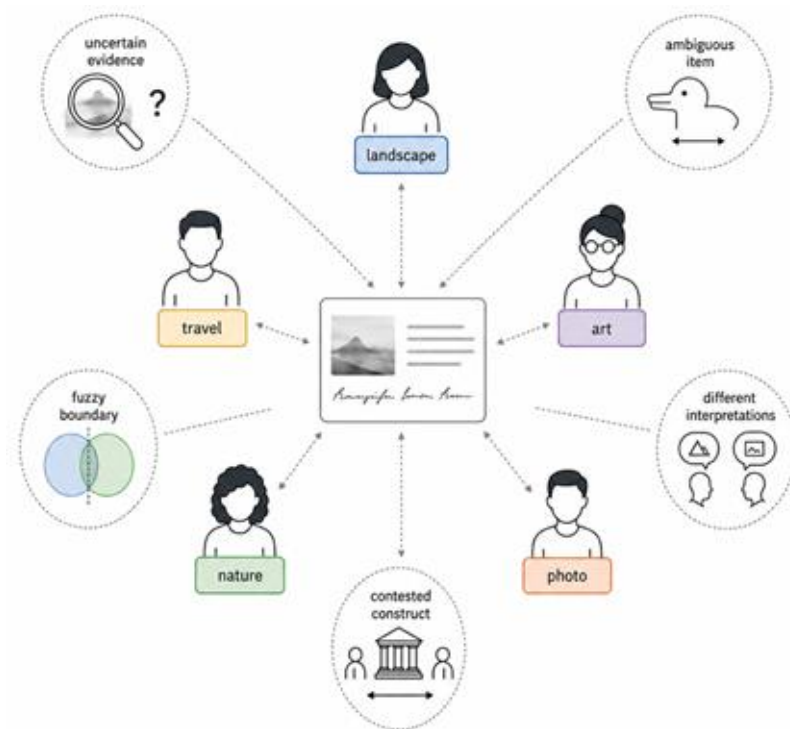
	urban_compact	urban_open	urban_rest
[1,]	0	0	12
[2,]	0	12	0
[3,]	1	11	0
[4,]	0	11	1
[5,]	3	9	0
[6,]	2	10	0
[7,]	12	0	0
[8,]	11	1	0
[9,]	9	3	0
[10,]	8	4	0

Human Label Variation

When people give different reasonable labels, the variation may reveal:

- uncertainty in the evidence
- ambiguity in the item
- a fuzzy category boundary
- different but relevant interpretations
- a contested social or institutional construct

→ **Disagreement can be a signal.**



What changes in evaluation?

Metrics that require one definitive label per item are **not meaningful**

- accuracy
- precision
- AUROC
- cross-entropy with hard labels

Evaluate against **label distributions**, not only majority labels

- cross-entropy with soft labels
- Brier score
- JS or KL divergence

Core shift:

Did the model match the ground-truth label?



Did the model capture the range and uncertainty of plausible human judgments?

Training on label distributions improves calibration



Standard approach: Multiple expert labels are collapsed into a majority label

$$(7,2,1) \rightarrow (1,0,0)$$

The model is trained with, e.g., categorical cross-entropy as though the selected class were unambiguously correct.

Distributional approach: Retain the empirical distribution of expert votes:

$$q = (0.7, 0.2, 0.1)$$

Train the model prediction by optimizing distributional metrics

Every class receiving annotations influences the loss.

→ **Discourages concentrated predictions. Improves calibration**

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Understanding Uncertainty in Data-Driven Research

Questioning Ground Truth Labels Active Learning

Unit 2



Labels are often the expensive part

Unlabeled data may be abundant.

But reliable labels may require:

- expert time
- laboratory tests
- careful reading or inspection
- access to sensitive information



Question: If we can label only 100 cases,
which 100 should we
choose?

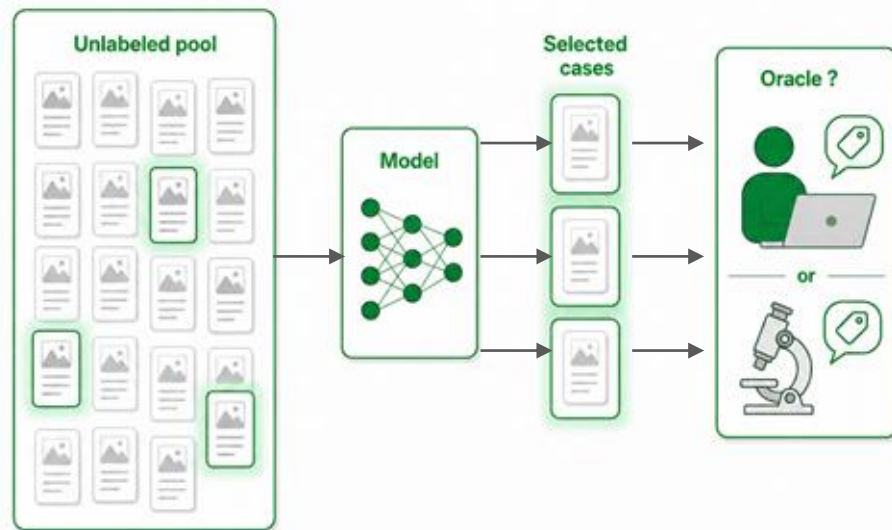
Active learning selects the next labeling requests



Active learning is an iterative process in which a model or selection rule chooses unlabeled examples for an oracle to label.

The **oracle** may be a person, expert panel, measurement, or experiment.

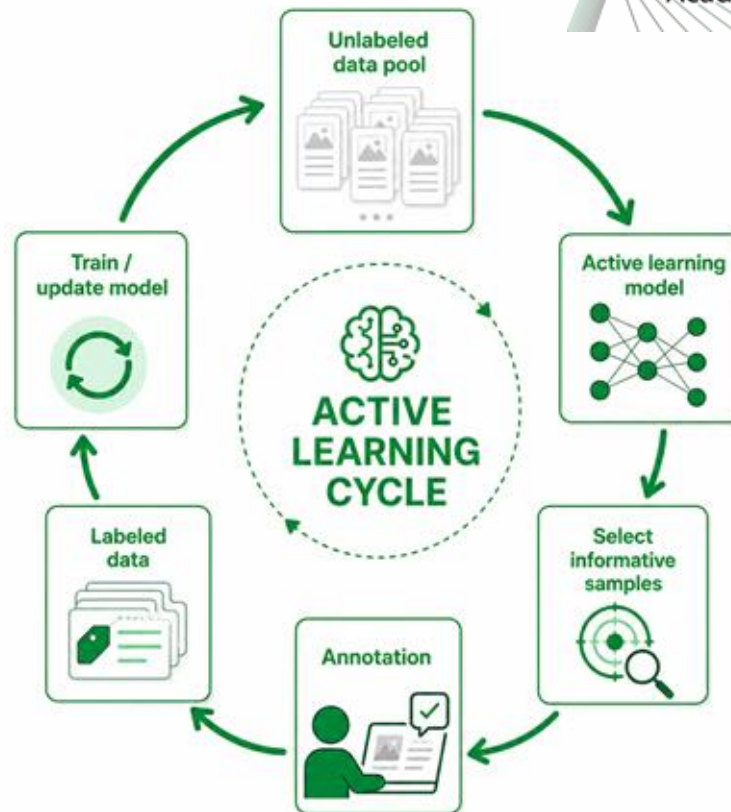
The goal is to reach useful performance with fewer labels.



The active learning loop

1. Begin with a small labeled set and a larger unlabeled pool
2. Train a model on the labeled set
3. Score unlabeled examples with a query strategy
4. Ask for labels for the selected batch
5. Add the new labels and repeat

Stop when the budget is exhausted or further labels add too little value.



Uncertainty sampling



For multiclass predictions, we can query examples with

- **smallest margin**: the top two classes are nearly tied

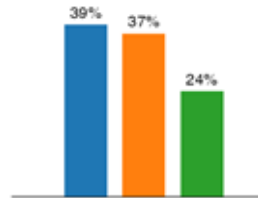
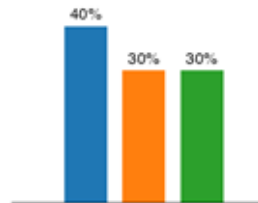
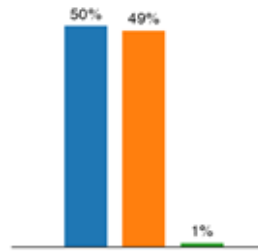
$$\alpha(x) = p(y_1|x) - p(y_2|x)$$

- **highest entropy**: probability is spread across several classes

$$\alpha(x) = - \sum_y p(y|x) \log p(y|x)$$

- **least confidence**: the top predicted class has low probability

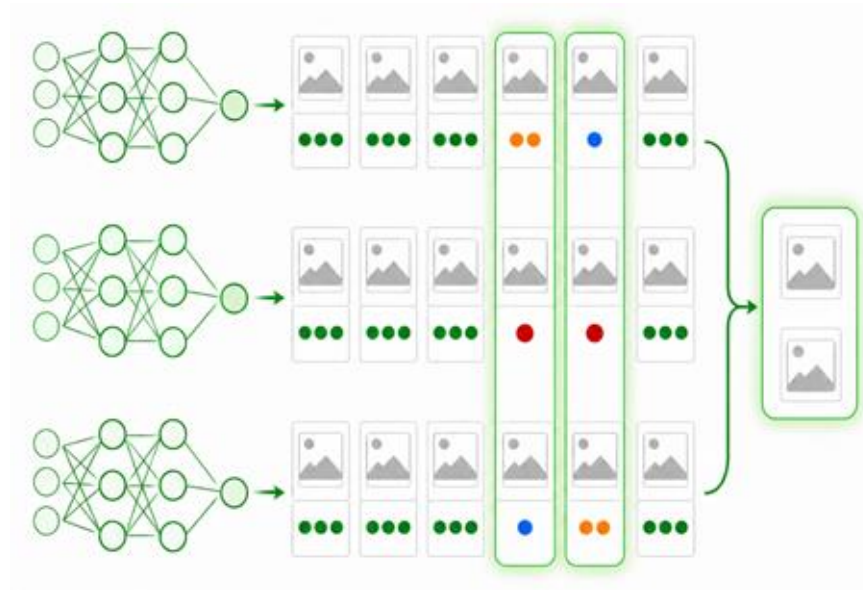
$$\alpha(x) = 1 - p(y_{\max}|x)$$



Query by committee:

1. Train several plausible models
2. Query examples on which their predictions differ most

Disagreement can reveal regions where the current labeled data do not determine one stable model.

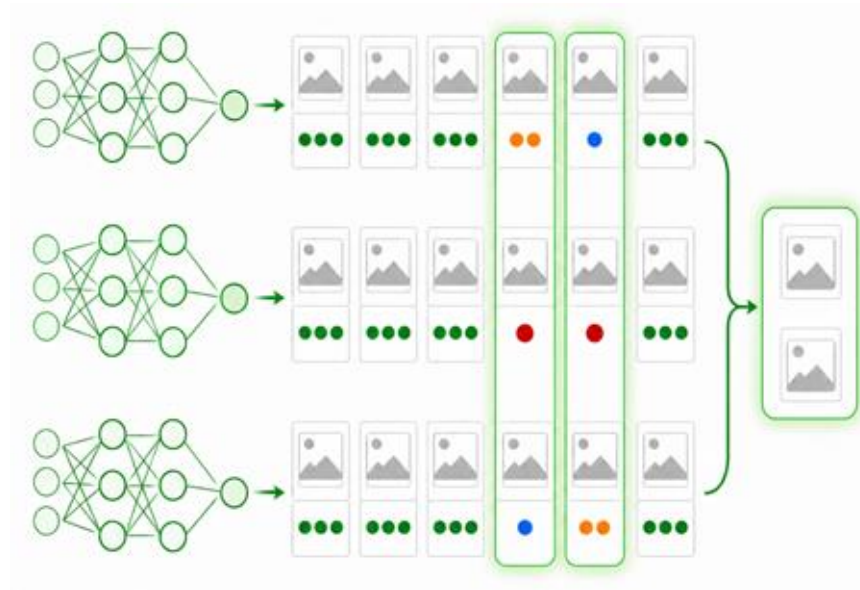


BALD: Bayesian Active Learning by Disagreement

BALD specifically searches for data points where individual committee members are

- highly **confident**
- but wildly **disagree with each other**

Labeling such a point can resolve this disagreement.



BALD: Bayesian Active Learning by Disagreement



Maximizing Mutual Information:

$$BALD = \underbrace{H[y|x, D]}_{\text{Overall uncertainty}} - \underbrace{E_{\theta}[H[y|x, \theta, D]]}_{\text{Average uncertainty of individual models}}$$

The overall uncertainty/entropy of the model's prediction for a given data point
→ **high when models disagree**

The average uncertainty of each individual model in the committee
→ **low when individual models certain**

H : entropy/uncertainty

y : prediction

x, D : data point

E_{θ} : average over models θ

Active sampling changes the labeled dataset



The actively labeled set is generally **not an independent random sample** of the population.

As a result:

- class frequencies may be distorted
- uncertain or unusual cases may be overrepresented
- ordinary accuracy estimates on that set may be misleading

Keep a separate, representative validation and test set.

Data usually
outlives
models.

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Understanding Uncertainty in Data-Driven Research

Questioning Ground Truth Labels Active Learning for Human Label Variation

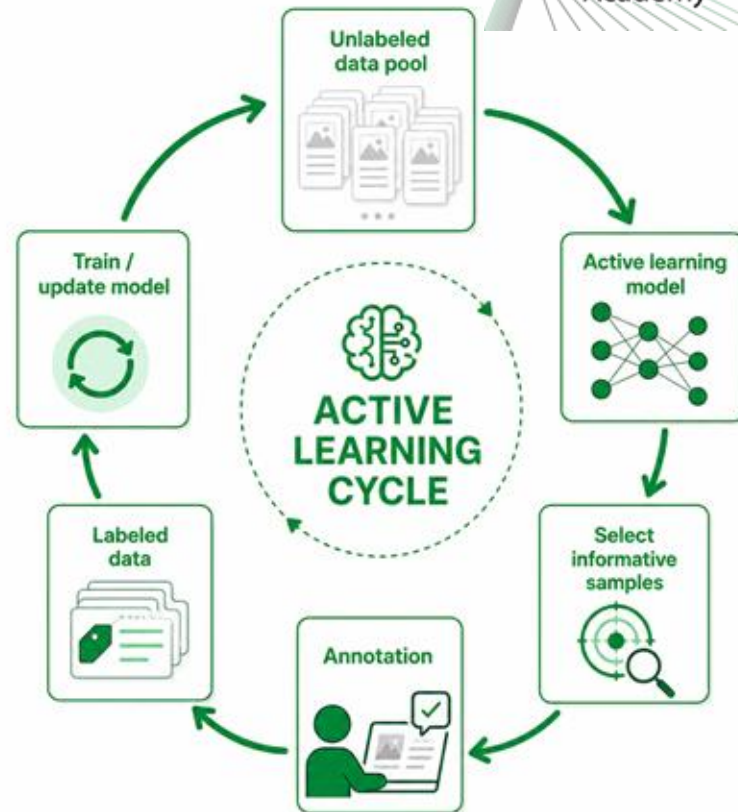
Unit 2



Active learning under human label variation

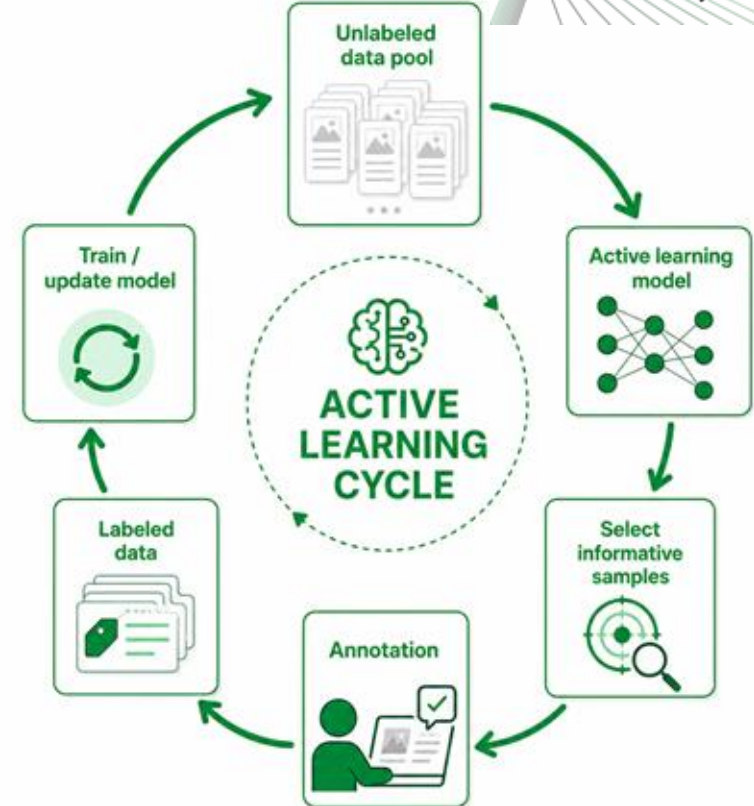
Assumptions in classical active learning:

1. There exists a single ground truth label per instance.
2. The oracle provides the ground truth labels without any noise.
3. The annotation difficulty or cost is equal for all instances.



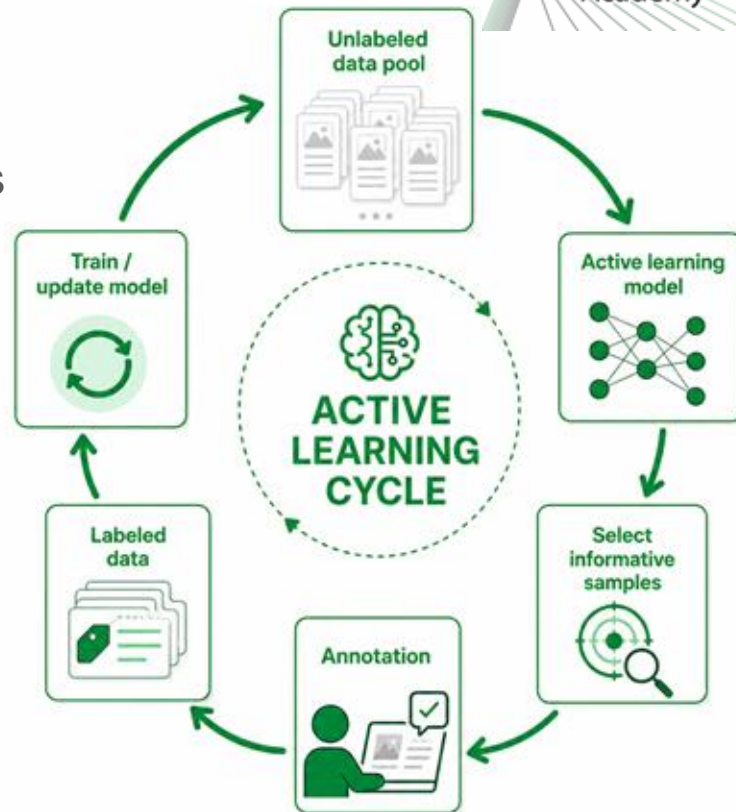
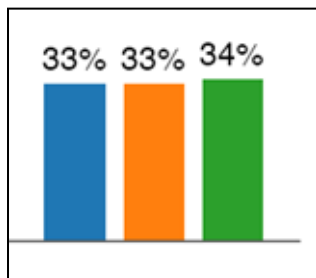
Active learning under human label variation

Distributional measures must be used for optimizing and evaluating the classifier.



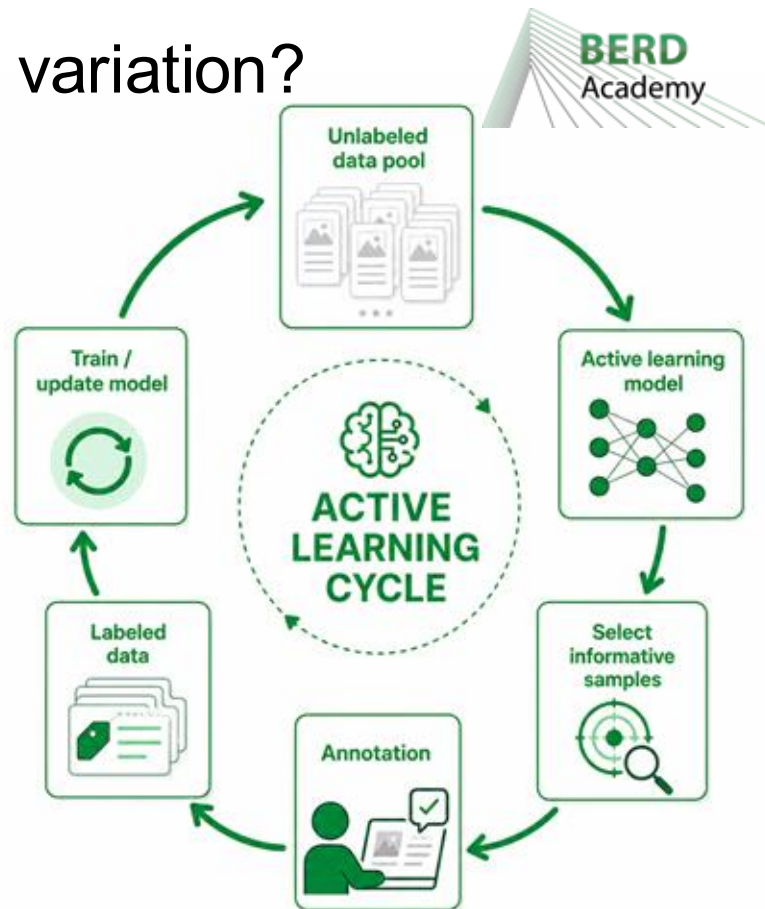
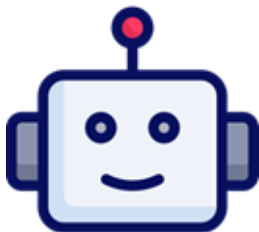
What changes under human label variation?

Classical informativeness or representativeness sampling are unsuitable, as they ignore the **option of labeling instances multiple times** and fail to process distributional labels.



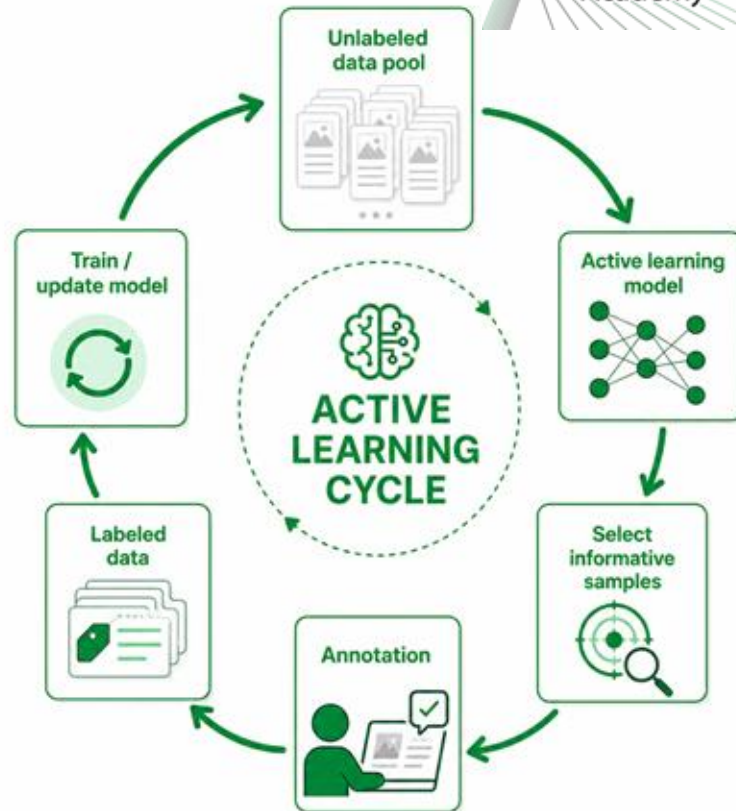
What changes under human label variation?

It matters who provides the label(s). An annotator acquisition function must decide not only whether to query a human or a language model, but also whether to obtain one or multiple annotations and from which specific annotator or model.



What changes under human label variation?

It is non-trivial to distinguish true label variation from noise, especially when labels can be sourced from both humans and language models, each with differing capabilities and output formats.



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Understanding Uncertainty in Data-Driven Research

Questioning Ground Truth Labels Natural Language Inference

Unit 2



Natural Language Inference

Given a context, a statement can be either

- Definitely correct (Entailment)
- Definitely incorrect (Contradiction)
- Neither (Neutral)

Context: A guitarist is playing in a band

Statement: Some people are performing

Label: Entailment

A black and white dog running
through shallow water

Two dogs running through water.

Entailment
($n_i = 42$)

Neutral
($n_i = 14$)

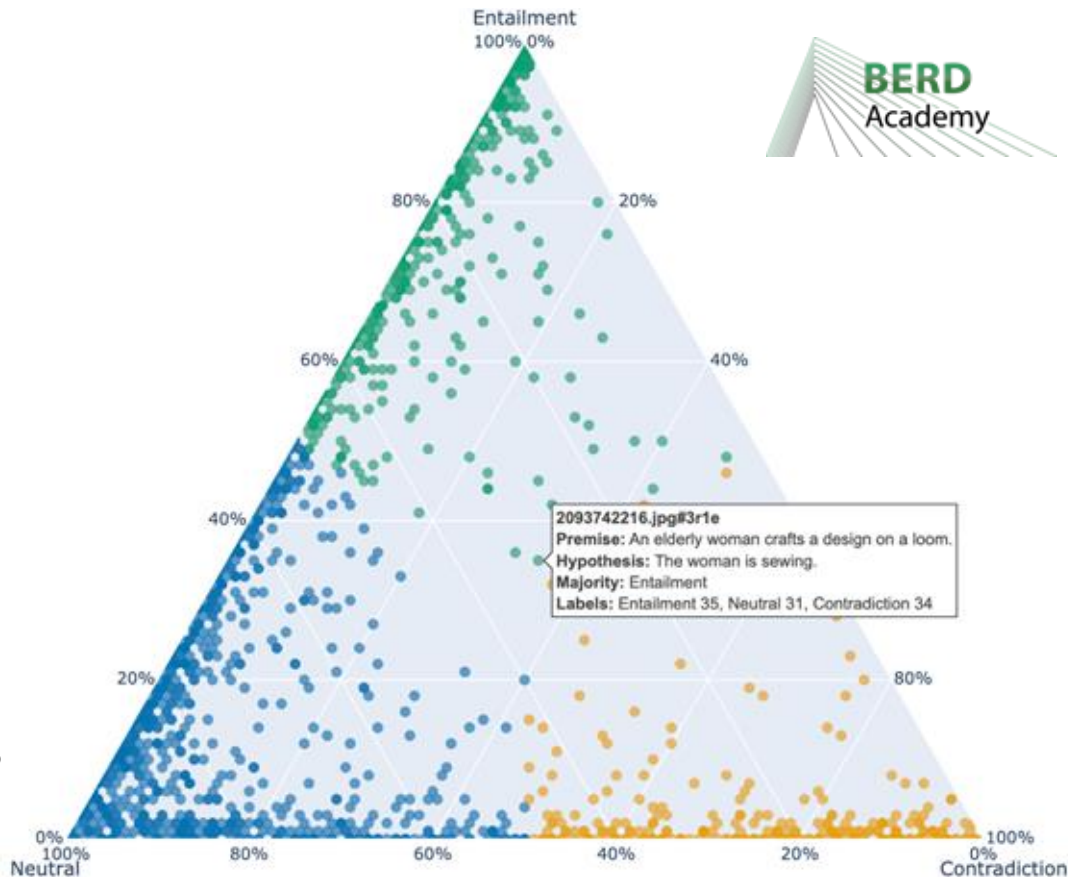
Contradiction
($n_i = 44$)



Homework Task

How to read ternary plots

- Scatter plot of the vote distribution of ChaosSNLI.
- Each point represents one instance.
- Its location is determined by the vote distribution.
- Corner points represent 100 votes for the respective class



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